MOONES BERTINA

Engineering Department, Tarbiat Modares University
Gmail | Academic mail | +98 (920) 301-1950 | LinkedIn | Skype

About Me

Civil engineer specializing in transportation planning and intelligent transportation systems. Research began with deep reinforcement learning (PPO) for single-intersection adaptive signal control, and has since extended to Multi-Agent Reinforcement Learning (MARL) for network-level, coordinated signal control, with hands-on experience in SUMO, VISSIM, and Visum. Growing interest in spatial data analysis for transportation, including geospatial network preparation and self-directed exposure to QGIS. Professional experience as a traffic analyst and structural engineer, bridging data-driven methods with real-world engineering practice.

Education

Tarbiat Modares University, Transportation Planning — Tehran, Iran
Master of Science (GPA: 17.80/20) — *Sep. 2025 – Present*

Isfahan University of Technology, Civil Engineering — Isfahan, Iran
Bachelor of Science (GPA: 3.1/4) — *Sep. 2019 – Sep. 2023*

Salam-Vanak High School — Tehran, Iran
Mathematics and Physics (High School Diploma, GPA: 4/4) — *Sep. 2016 – Jun. 2019*

Research Interests

Traffic Safety | Deep Reinforcement Learning | Intelligent Transportation Systems | Adaptive Traffic Signal Control | Geospatial Analysis in Transportation | Smart Cities

Selected Courses

- System Analysis
- Transport Planning
- Transportation Demand Analysis
- Airport Design and Planning (Master's)
- Machine Learning
- Computer Programming (Python, R, C, C++) (Bachelor)
- Statistics and Probability (Bachelor)

Research Experiences

Research Assistant — 2026 – Present
Transportation Lab – Part-time — *Tarbiat Modares University, Tehran, Iran*

- Extending prior single-intersection PPO-based signal control work (see Seminar Project below) to network demand modeling and Multi-Agent Reinforcement Learning (MARL)-based signal control, with a focus on emergent green wave coordination.

Professional Experiences

Traffic Analyst — 2024 – 2025
EttehadRah – Full-time — *Tehran, Iran*

- A consulting company for transportation planning and traffic studies.
- Conducted traffic data collection, processing, and analysis to support transportation planning and traffic-study deliverables for client projects.
- Worked with transportation network and demand-modeling data, including preparation of spatial/GIS-based network inputs for simulation and analysis tools (e.g., Visum).
- Contributed to traffic studies involving intersection and corridor performance evaluation, supporting data-driven recommendations for network improvements.

Structural Engineer and Designer — Dec. 2023 – Feb. 2025
Atishar – Full-time — *Tehran, Iran*

- A consulting company for Construction and Architecture.
- **Full-cycle structural and architectural project design (ETABS, AutoCAD, SAP/SAFE), including structural analysis, cost estimation automation with Python, and site/landscape planning.**

Technical & Research Experiences

Seminar Project

Design and Implementation of an Adaptive Traffic Signal Controller Based on Deep Reinforcement Learning (PPO) at a Single Intersection Using the SUMO Simulator

- Developed and implemented a traffic signal control system using Proximal Policy Optimization (PPO), a deep reinforcement learning algorithm, to adaptively manage traffic flow at a single intersection.
- The simulation environment was built using SUMO (Simulation of Urban MObility), enabling realistic traffic modeling and evaluation of the proposed controller's performance under various traffic conditions.
- This single-agent formulation laid the foundation for current MSc research on network-level, multi-agent signal coordination (see Research Experiences above).

Selected Projects

1. **Transportation Demand Modeling Project — PTV Visum:** Built and calibrated a 5-zone transportation network model for Tehran using PTV Visum, including OD matrix calibration via the gravity model (KALIBRI). Performed modal split analysis using a logit model and conducted PrT/PuT (private/public transport) assignment. Produced a comprehensive analytical report and published the project on GitHub. (MSc. 2nd Semester 2025–2026)
2. **Road Construction Project:** Designed a roadway network using Civil 3D and AutoCAD, considering alignment, grading, and material specifications for construction. (Grade: 19.5/20, BSc. 2nd Semester 2021–2022)

Selected Project

Traffic Flow Analysis (2nd Semester 2021–2022, Bachelor's Degree), Isfahan University of Technology

- Conducted a study on local traffic flow using manual counting methods during peak hours at a busy intersection on campus.
- Analyzed collected data using Python, identifying congestion patterns and causes.
- Proposed solutions for improving traffic conditions by implementing traffic signals with optimized peak-hour timing.

Skills

- **Programming Languages:** Python, R, C/C++, HTML, CSS
- **Libraries & Frameworks:** PyTorch, TensorFlow, RLlib (Ray), SUMO-RL, NumPy, Pandas
- **Tools & Software:** ETABS, Civil3D, AutoCAD, SAFE, WaterGEMS, SewerGEMS, SUMO, Visum, VISSIM, QGIS (basic/self-taught)
- **Document Creation:** Microsoft Office Suite, LaTeX, Adobe Suite

Standardized Test Scores

GRE General: 303 — *Oct. 2024*

Quantitative: 163, Verbal: 140, Analytical Writing: 3.0

IELTS: 7 Overall — *Feb. 2023*

Reading: 6, Listening: 8, Speaking: 7, Writing: 6